

Predicting Treatment and Functional Response in Macular Hole Using Generative Adversarial Networks (GANs)

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Abstract: Macular holes, a critical retinal illness that leads to central vision impairment, it requires successful surgical treatment and precise prognostic models. This research explores the capability of deep learning models in macular-hole closure prediction after surgery by creating and comparing two Generative Adversarial Networks (GAN) architectures: a custom cGAN (Conditional GAN) and a modified Pix2Pix model. These models were trained on a pre and post-treatment OCT image dataset that was optimized to learn the intricate relationship between pre and post treatment OCT images. Both the models showed promising performance in generating synthetic post-treatment images which closely matched the actual scans. The cGAN implemented specifically for this task had an SSIM of 0.662, PSNR of 28.227 and LPIPS of 0.312. Likewise the modified Pix2Pix model provided an SSIM of 0.554, a PSNR of 26.854 and a LPIPS of 0.149. Visual analysis of the resulting images also supported the capabilities of the models. The findings indicate that the models were able to effectively capture subtle morphological changes brought about by treatment, highlighting their potential clinical applications. Such GAN-based models may assist clinicians in preoperative planning, choice of surgical strategy, and patient management by offering a useful tool for macular hole closure prediction.

Keywords: Optical Coherence Tomography (OCT), Macular Hole (MH), Generative Adversarial Networks (GANs), Image-to-Image Conversion, Conditional GAN (cGAN), Pix2Pix Model, Forecasting Surgical Results.

1. Introduction

Retinal diseases are a main cause of vision impairment and blindness worldwide [1]. Retina, a light-sensitive membrane that covers the internal surface of the eye, having a vital role in vision. The macula is the central part of the retina which is responsible for central vision, reading vision and colour vision. A macular hole (MH) is a full thickness defect in the central point of the macula which leads to immediate and severe vision impairment along with significant decline in the quality of life [2,3]. While age-related alterations in the vitreous humor and the traction on the retina resulting thereof are believed to be the genesis of macular holes [4], there are also other risk factors that include myopia, detachment of the retina, and ocular trauma [2]. Clinically, macular holes present with scotomas, metamorphopsia, and progressive loss of vision [5], vitrectomy, which combines internal limiting membrane peeling and gas, remains the primary treatment for macular holes. However, the anatomical as well as the visual outcomes after surgical treatment often show significant variability [6]. A simple non-invasive test known as optical coherence tomography (OCT) is done to diagnose MH which is seen as a full thickness defect in the retinal layers. The OCT is also an effective tool to monitor the outcomes of surgery. A successful outcome will result in closure of the gap in the retinal layers with restoration of normal contour while a failed surgery will show persistent gap in the retinal layers along with some separation of the neurosensory retina. This variability calls for highly improved predictive models that can project treatment effectiveness and advise on the best mode of management in patients. With the current advancements in the domain of artificial intelligence, the use of generative adversarial networks have resulted in significant change in

medical image analysis [7]. GANs work by a competing scheme of training with two neural networks—one for generating new data, and another that is trained to differentiate the authenticity of this generated data against the real samples [8]; it is because of this dynamic interplay that GANs are efficiently learned with complex data patterns, therefore of much value in medical imaging applications, such as in the case of retinal disease analysis [9]. This research paper is focused on applying GAN technology to predict the results of macular hole closure based on OCT images captured before and after treatment. This paper investigates and provides a comprehensive analysis of the performance of multiple models for prediction: a modified Pix2Pix-[10] based model, and a custom cGAN [11]. The evaluation extends beyond visual inspection, employing quantitative metrics such as the Structural Similarity Index (SSIM) [12], Peak Signal-to-Noise Ratio (PSNR) [13], LPIPS [14], MAE [15], MSE [16], in order to evaluate the quality of generated images. The models were learned from a carefully pre-processed OCT dataset to encode the subtle morphological changes of the treatment process, with the intent to create a strong predictive model that improves patient care and informs treatment choices for patients with macular hole. In this paper, section 2 summarizes existing research methodology relating to macular hole and surgery. Section 3 explains the dataset and the suggested approach, section 4 outlines the criteria used for evaluation, section 5 discusses the results of experiments and evaluates the proposed methodology and lastly, section 6 offers discussions and conclusions.

2. Literature Review

This section seeks to discuss the latest developments in using generative adversarial networks (GANs) to improve diagnostic and predictive functions, specifically focusing on macular hole surgery. Optical Coherence Tomography (OCT) has emerged as a valuable tool in contemporary ophthalmology and has found applications in other areas, thanks to its ability to

provide high-resolution, in vivo images (Kalupahana et al., 2024). OCT image analysis, nonetheless, conventionally requires significant expertise, and may be a time consuming and subjective process. In light of these shortcomings, scientists are increasingly exploring deep learning (DL) techniques to automate and enhance OCT image analysis processes [17] [18]. Deep learning has the possibility of augmenting diagnostic tests with increased efficiency and less variability. To put this in perspective, Kalupahana et al. (2024) discussed the utilization of deep learning (DL) integration with Optical Coherence Tomography (OCT) for detecting ophthalmic disease, highlighted by the utilization of DL models to classify diseases like Age-Related Macular Degeneration (AMD). Since age-related macular degeneration is common macular diseases, this article discusses the implementation and use case of artificial intelligence (AI), such as machine learning (ML), neural networks architectures (NN), and also deep learning (DL) to improve predictive and diagnostic functionalities. Han et al, has developed a model utilizing OCT images for predicting whether anti-VEGF injections can result in gains. They have created an AI based model using the DenseNet201 to forecast anatomical improvement in neovascular AMD (nAMD) after anti-VEGF treatment, from optical coherence tomography (OCT) images. The model's predictive performance improvised after the of including OCT images captured after the first and second anti-VEGF injections [19].

The following sections will present an analysis of existing literature, emphasizing recent progress in the use of DL specifically focusing on the OCT imaging with potential to improve the specificity of macular hole outcome predictions, as presented in Table 1. Findings indicate that the addition of artificial intelligence (AI) and deep learning techniques can significantly enhance both preoperative assessments and postoperative predictions. Research conducted by [1] introduced a conditional variational autoencoder (VAE) that effectively predicted postoperative optical coherence tomography (OCT) images following MH surgery. Utilizing a dataset of 150 patients, the authors implemented a preprocessing pipeline that included

automated registration and data augmentation, achieving a notable accuracy of 94.7% in predicting neurosensory retinal areas. This study outlines the potential of variational autoencoders in generating clinically useful OCT images that can aid surgeons in evaluating surgical outcomes. In another study, [2] aimed at predicting treatment results in neovascular AMD employing an attention-steered GAN. Through analysis of a database containing 1,684 OCT images, their model outperformed predictions made by human graders of residual retinal fluid after treatment. This highlights the ability of GANs to detect subtle changes in retinal structure affected by various treatment drugs, indicating the potential of using GANs in tailoring therapy for patients with AMD. Authors in [3] extended this work further by using a conditional GAN (cGAN) to synthesize post-treatment OCT images from baseline information in neovascular AMD. The integration of fluorescein angiography & indocyanine green angiography images also helped improve predictive accuracy, demonstrating the positive synergy between different imaging modalities and AI models. Another study extended the application of GANs by utilizing the pix2pixHD framework to forecast short-term responses to anti-vascular endothelial growth factor (anti-VEGF) therapies [4]. Their approach had a high accuracy rate in producing clinically interpretable fake OCT images, ensuring the feasibility of GANs for providing vital information for clinical decision-making. The uses of AI and deep learning go beyond single disease predictions. Author in another study presented a comprehensive overview of AI use in ophthalmology, with a emphasize on the highlighting the shift from traditional machine learning to deep learning methods for diagnosis and management of several retinal diseases [5]. Their conclusion stressed the need for large, heterogeneous datasets to improve model performance and overcome data scarcity issues. Additionally, the authors in [6] proved the successful generation of realistic OCT images based on GANs that had the potential to enhance training datasets and enhance the robustness of diagnostic algorithms. This finding is in accordance with the findings of [7], who presented various GAN

architectures and their capability to produce high-resolution retinal images and hence enable better training of deep learning models.

In addition, researchers investigated machine learning models for anatomical outcome prediction following idiopathic macular hole surgery with impressive results from random forest models [8]. The study highlighted the need to combine clinical characteristics with imaging information in predictive models. While encouraging findings exist, various challenges prevent AI research from being translated into clinical practice seamlessly. Incorporating generative models into standard workflows demands rigorous validation and standardization of methods. Studies by [9] and [20] emphasize the continued imperative for dependable measures to test AI performance but also resolve ethical concerns as well as data security. Overall, the use of generative deep learning models in ophthalmology, such as in macular hole surgery and AMD management, presents immense prospects for enhancing diagnostic precision and clinical decision-making. While present studies illustrate significant advances, continuous efforts are required to resolve challenges to maximize the utilization of AI technology in clinical practice. To further highlight the necessity for meticulous surgical evaluation of macular holes using OCT, it is also important to recognize possible complications that may occur despite successful closure of macular holes. Ye et al. (2023) performed a spectral domain OCT investigation of post-vitrectomy gliosis in highly myopic macular holes and reported that inverted ILM flap surgical techniques enhanced closure rates but at the cost of increased risk of undue gliosis, particularly in eyes with extended axial lengths. In light of these potential surgical issues, it is necessary to emphasize the potential complications that can arise after a successful surgery of a macular hole [21].

Table 1. Literature Review concise in a tabular form

Authors	Year	Methods	Clinical Entity	Results
Kwon et al. [1]	2024	Conditional variational autoencoder (VAE)	Macular hole surgery	94.7% accuracy in predicting postoperative neurosensory retinal areas from OCT images.
Moon et al. [2]	2023	Attention-guided GAN	Neovascular AMD	Outperformed human examiners in predicting residual retinal fluid post-treatment.
Lee et al. [3]	2021	Conditional GAN (cGAN) with fluorescein & indocyanine green angiography	Neovascular AMD	Generated post-treatment OCT images from baseline data, improving predictive accuracy.
Liu et al. [4]	2020	pix2pixHD GAN framework	Anti-VEGF treatment in neovascular AMD	High accuracy in generating synthetic OCT images for short-term response prediction.
Jin & Ye et al. [5]	2022	Review of AI/deep learning in ophthalmology	General retinal conditions	Highlighted the shift from conventional ML to deep learning for diagnosis and management.
Odaibo et al. [6]	2019	GANs for OCT synthesis	Retinal imaging (general)	Demonstrated realistic OCT image synthesis to improve training datasets.
Wang et al..[7]	2021	Review of GAN architectures	Retinal imaging (general)	Discussed GANs' potential to generate high-quality retinal images for improved model training.
Xiao et al. [8]	2021	Random forest models	Idiopathic macular hole surgery	Achieved robust predictions of anatomical outcomes by integrating clinical and imaging data.
Ferro Desideri et al. [9]	2023	AI-inferred large language models (LLMs)	Age-related macular degeneration (AMD)	Emphasized the need for validation and ethical considerations in AI applications.
Kalup hana et. al. [17]	2024	Analysis of various deep learning methods.	Applications of Optical Coherence Tomography across multiple disciplines	Provides a review of current deep learning applications within multidisciplinary OCT.

Moraru et al. [18]	2020	Review of AI and Deep Learning techniques in ophthalmology	Retinal diseases/General ophthalmology	AI has widely demonstrated reliability in detecting retinal diseases and also in enhancing diagnostic accuracy for conditions affecting the eye's posterior segment.
Han et al. [19]	2024	Deep learning-based algorithm, specifically DenseNet201	Neovascular Age-Related Macular Degeneration (nAMD)	AI model surpassed ophthalmologists in predicting complete remission following three intravitreal anti-VEGF injections
Xu et al. [20]	2021	AI models for OCT prediction	Central serous chorioretinopathy	Predicted post-therapeutic visual acuity and OCT images using AI.
Ye et al. [21]	2023	Retrospective observational study; SD-OCT analysis: ILM peeling versus the Inverted ILM flap approach	Highly Myopic Macular Hole (HMMH) and Idiopathic Macular Hole (IMH)	Inverted ILM flap is better for HMMH closure but linked to higher excessive gliosis, especially with axial length >29.985mm. All IMHs closed.

3. Methodology

The flowchart, illustrated in Fig 1, outlines a research study investigating deep learning for predicting macular hole closure post-surgery. Beginning with an OCT image dataset, the process involves preprocessing and standardization before applying a generative model approach. Two distinct GAN architectures are explored: Approach 1: a custom conditional GAN (cGAN) and Approach 2: the modified pix2pix model. Both undergo adversarial training. Performance is evaluated using metrics (SSIM, PSNR, MAE/MSE, LPIPS) and ophthalmological validation. A comparative analysis determines the more effective architecture

for predicting macular hole closure. This section explains each step involved in predicting the macular hole closure post-surgery images.

3.1 Dataset Preparation

The dataset, provided by Sankara Nethralaya, Chennai, Tamil Nadu, included images from 74 patients who underwent surgery for macular holes. A standard surgical procedure involving three-port pars plana vitrectomy (23 or 25 G) followed by staining with Brilliant blue dye and Internal limiting membrane peeling followed by gas tamponade was conducted for all the patients. All were subjected to pre-operative OCT scans. The OCT was repeated 6 weeks after the surgery to look for MH closure. All the scans were done using the Cirrus HD-OCT Angioplex 5000 system (Carl Zeiss Meditech, Inc, Dublin, CA, USA, Ver 11.0.0.29946), for each patient, 128 OCT slices were available. This dataset is not publicly available and was crucial for analysing the pre-and post-operative morphological changes related to the treatment. The specific steps undertaken in the preparation of the dataset are as follows:

- o **Image Data Collection and Preparation:** The OCT scan image passing through the fovea was selected for each patient. Besides this central image, one, two or three images per patient were collected both for the pre and post-surgery conditions, depending on the variation of the scans. All OCT images were cropped and resized to measure 256×256 pixels in dimension; this dimension is constant throughout the entire dataset to ensure model consistency for neural networks.
- o **Augmentation(s) of data:** The augmentation techniques are applied to the training dataset, so the model is robust and does not overfit. The following are augmentations: Original Image, Horizontal Flip, Vertical Flip, Rotation by 90, Rotation by 180, Rotation by 270, adjustments to brightening (scaling brightness to 0.5 and 1.5).

- o **Dataset-split and Final-dataset:** The dataset was divided into two parts: training and testing. Before applying the augmentation(s) 70/30 split was used. That entailed 70% of the data should be applied to the train dataset and 30% was set-aside for the testing set. In other words, after augmenting the training set the final dataset has total training images: 1032 image pairs (including augmentations), and total testing images: 56 image pairs.

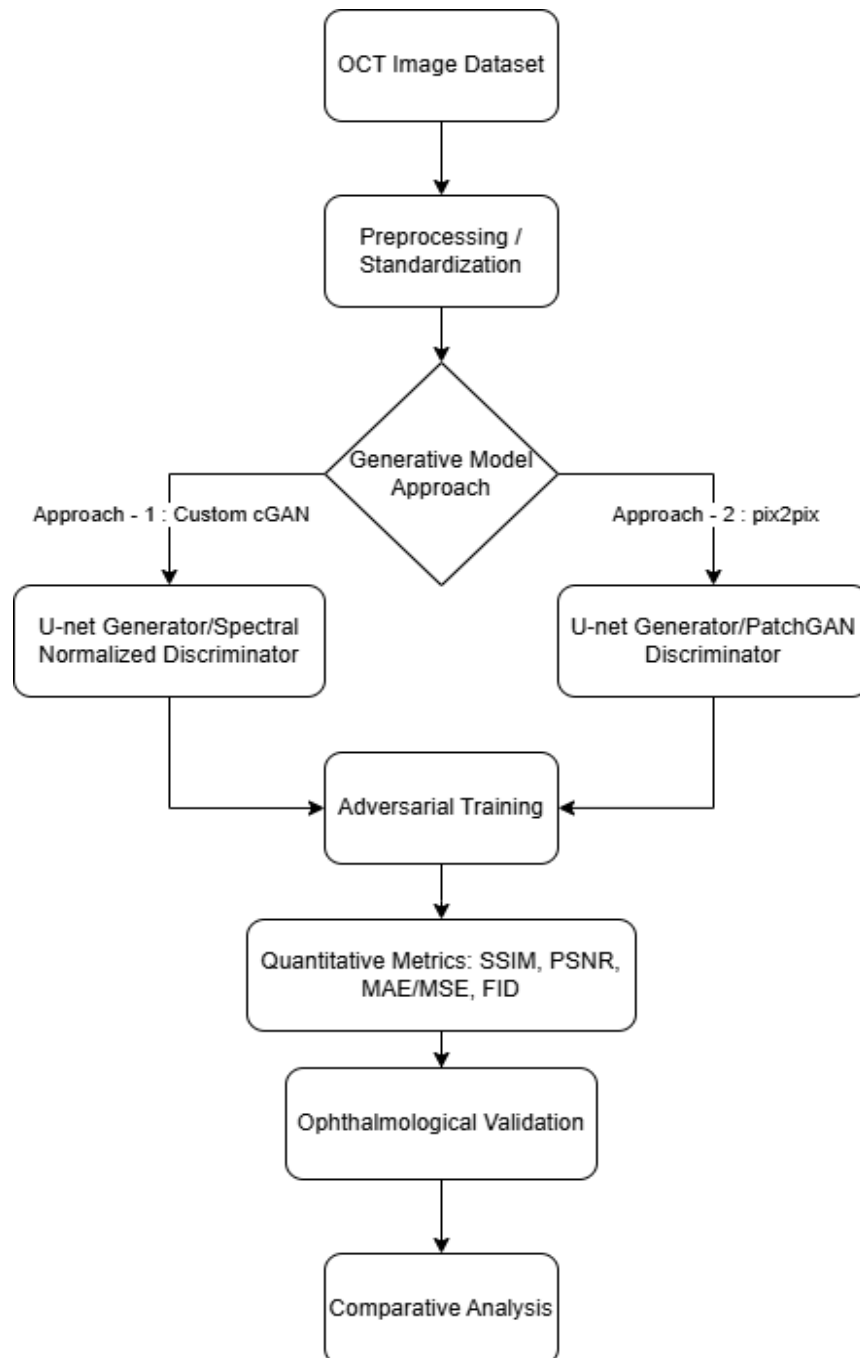


Fig.1 : Flowchart illustrating the development and comparison of two cGAN architectures (custom cGAN and modified pix2pix) for predicting macular hole closure post-surgery using OCT images.

3.2 Generative Model Approach: This is the core of the process which explore the two different GAN-based approaches, training models to generate realistic OCT images similar to the training data.

A. Approach-1: Custom Conditional Generative Adversarial Network (cGAN):

This approach utilizes a custom Conditional Generative Adversarial Network (cGAN) developed for image-to-image translation specifically predicting post-operative OCT images from pre-operative ones. The cGAN architecture comprises a generator and a discriminator, both tailored for this task.

1. Generator: The generator employs a U-Net-like architecture with skip connections to effectively capture both high-level and low-level features, crucial for preserving anatomical details in the generated post-operative images. The input for the generator is a pre-operative OCT image. The architecture consists of the following key components:
 - a) Downsampling Path: A series of convolutional blocks progressively downsample the input image. Each block typically consists of a convolutional layer (Kernel size is equivalent to 4, stride is 2, and the padding value set to 1), a Leaky ReLU activation function for non-linearity, and Instance Normalization to improve training stability and performance. These downsampling layers extract progressively abstract features from the input image, expanding the feature channels from 64 to 512.

- b) **Bottleneck Layer:** The bottleneck layer, situated at the deepest part of the U-Net, further processes the encoded features. This layer is essential for optimizing information content prior to initiating the upsampling process.
 - c) **Upsampling Path:** Mirroring the downsampling path, the upsampling path employs transposed convolutions (also called deconvolutions) to incrementally increase the spatial resolution of the feature maps. The skip connections link the corresponding layers from the downsampling and upsampling paths, are essential for transferring fine-grained details from the earlier layers to the later layers, helping to preserve sharp edges and textures in the generated image.
 - d) **Output Layer:** The final layer present in the generator is a transposed convolution followed by a Tanh activation function which produces the predicted post-operative OCT image. The Tanh activation which makes make sure that the output pixel values are within the range of -1 to 1.
2. **Discriminator:** The function of the discriminator is to separate between real post-operative images and fake post-operative images generated by the generator. It uses a PatchGAN architecture with spectral normalization [22] for enhanced training stability. The discriminator architecture includes:
- a) **Convolutional Layers:** The initial convolutional layers present are used to extract features from input image using a kernel size is equivalent to 4, a stride of 2, also padding value set to 1, spectral normalization, and a LeakyReLU activation ($\alpha=0.2$). The number of feature channels progressively expand from 32 to 512. Crucially, this network is conditioned on both real pre-operative and

real post-operative images, which are provided as a pair to guide the generation of synthetic post-operative images.

- b) Spectral Normalization Blocks: Subsequent convolutional blocks incorporate spectral normalization. Spectral normalization is a crucial technique for stabilizing training of GAN by controlling the Lipschitz constant of the discriminator, preventing it from becoming too powerful and thus hindering the generator's learning.
- c) Output Layer: The final layer of the discriminator outputs a single scalar value; this value represents the discriminator's prediction of whether the input image pair (real pre-operative and generated post-operative) is real or fake. A sigmoid activation function ensures that the output produced has a probability between 0 and 1.
- d) Conditional Input: It's very important to emphasize that this is a conditional GAN. The discriminator obtains both the pre-operative image and the post-operative image (either real or generated) as input. This conditioning is what allows the GAN to learn the mapping between pre-operative and post-operative states.

Loss Function: The training of the custom cGAN model is driven by a composite loss function. The generator loss comprises two components: (1) Adversarial Loss: Implemented using Binary Cross Entropy (BCE), which encourages the generator to produce images that can fool the discriminator. (2) Reconstruction (L1) Loss: This loss actually helps to measures the pixel-wise difference between the generated and real post-operative images.

Training procedure: The model was trained using an adversarial framework, we employed the Adam optimizer for both the generator and discriminator. The loss function combined

Binary Cross Entropy (BCE) for discriminator and L1 Loss for reconstruction loss (alpha set to 100). The training was conducted for up to 200 epochs as further there was no improvement seen.

B. Approach-2: Modified Pix2Pix GANs:

This approach employs a modified Pix2Pix architecture—a well-established Conditional Generative Adversarial Network (cGAN) framework—for image-to-image translation. Specifically, it aims to generate post-operative OCT images from pre-operative OCT scans. The modified Pix2Pix model also consists of two main components: a generator and a discriminator.

1. **Generator:** The core architecture of the generator is based on the U-Net model with skip connections [23]. The U-Net incorporates the downsampling path (encoder) and an upsampling path (decoder). In the encoder, convolutional layers with a kernel size of 4, a stride of 2, and a padding of 1 are used, with Leaky ReLU activations (slope 0.2) and Instance Normalization applied (except in the first layer). The number of filters begin from 64 and gets doubles at each subsequent layer, take value up to 512 at the bottleneck. In the decoder, transposed convolutions (also with kernel size 4, stride 2, and padding 1) are used to restore the spatial resolution, while the corresponding encoder features are concatenated via skip connections to preserve fine details. The final output layer takes into consideration a Tanh activation function to normalize pixel values equivalent to the range which occurs between $[-1, 1]$.
2. **Discriminator:** The PatchGAN discriminator classifies overlapping patches rather than the entire image. It receives the combined input of the concatenation of the pre-

operative OCT image and corresponding post-operative image (real or generated), forming a 6-channel input for RGB images. The discriminator employs several convolutional layers (with kernel size 4, stride 2 except for the final layer which uses stride 1 and padding of 1), Leaky ReLU activations (slope is set to 0.2), and Instance Normalization (applied in all layers except the first). The output is a feature map of roughly 30×30 patches (for a 256×256 input), with each patch classified as real or fake.

Loss Function: The Pix2Pix model training employs a combination of losses: The adversarial loss using BCE With Logits Loss guides the generator in creating images that can deceive the discriminator, while the reconstruction (L1) loss evaluates the pixel-level disparity between the generated and actual post-operative images, weighted by a factor of 100; Perceptual Loss ensures that the generated images capture high-level features and textures by computing a loss using features obtained from a pre-trained VGG19 network (from its conv4_2 layer), further constraining the generator to match the semantic content of the real images; and Feature Matching Loss ensure the generator to produce images that yield similar multi-scale representations as real images by computing a loss between intermediate feature maps extracted from the PatchGAN discriminator for real and generated image pairs, which helps to reduce the discrepancy between the distributions of real and generated images.

Training procedure: The model is trained using an adversarial framework over 300 epochs having batch size set equivalent to 4. Both the generator and discriminator are optimized using the Adam optimizer (learning rate = 0.0002, betas = (0.5, 0.999)). Additionally, a learning rate scheduler reduces the learning rate by half every 100 epochs to facilitate finer updates later in training. During training, evaluation metrics such as PSNR, MAE, SSIM and LPIPS are

monitored on the test set (comprising 56 images). For qualitative assessment, sample outputs—including the input, generated, and ground truth images—are saved for visual inspection.

4. Evaluation metrics

The performance of both approaches is evaluated using several quantitative metrics:

4.1. Structural Similarity Index (SSIM)

The Structural Similarity Index (SSIM) is a perceived metric that measures the similarity between two images. In contrast to conventional metrics that only account for pixel-wise differences, SSIM accounts for structural information in changes, luminance, and contrast in order to make a more comprehensive judgment of image quality. It contrasts local luminance-normalized and contrast-normalized patterns of pixel intensities and is sensitive to structural distortions that are relevant to human vision. The SSIM value between -1 and 1 exists, with close to 1 represents a structural match, 0 for lack of similarity, and close to -1 for dissimilarity.

4.2. Peak Signal-to-Noise Ratio (PSNR)

The Peak Signal-to-Noise Ratio (PSNR) is a used measure to determine the quality of reconstructed images, specifically in the case of image compression and image denoising. It estimates the ratio of the maximum power possible for a signal (image) and the power of noise; it is represented in decibels (dB). An increase in the values corresponds to higher quality images while decreasing in the signal's poor image quality.

4.3. Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) is an evaluation metric that computes the average absolute error between corresponding pixels in the source and output images. It calculates how close the predictions are to the true values, lower values for better performance.

4.4. Mean Squared Error (MSE)

The Mean Squared Error (MSE) is also a simple evaluation metric that calculates the average of squared differences between the original and synthetic image pixels. It is calculated by squaring the difference obtained between each corresponding pixel in the original and synthetic images; added up for all these squared differences and divided by the total number of pixels. Smaller values represent better quality images with a perfectly reconstructed image resulting in an MSE of zero.

4.5. Learned Perceptual Image Patch Similarity (LPIPS) Score

The Learned Perceptual Image Patch Similarity score (LPIPS) is a measure that determines the perceptual similarity which occurs between images which calculates how visually similar two images are by comparing their feature representations which are derived from a pre-trained neural network. Unlike other measures, LPIPS measures the perceptual differences by analyzing differences in high-level feature activations and not pixel-wise differences. This method is more consistent with human visual perception and therefore, it is a strong metric to assess image quality and fidelity in image generation and style transfer tasks. Lower values correspond to higher perceptual similarity between the compared images.

5. Results and Discussion

This section debates about the results obtained from testing the two methodologies for post-operative OCT image generation from pre-operative scans: a custom Conditional Generative Adversarial Networks (cGAN) and a modified Pix2Pix model. The performances of both these models were tested using quantitative measures of different nature, including Structural Similarity Index (SSIM), Mean Squared Error (MSE), Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR) and Learned Perceptual Image Patch Similarity (LPIPS). The

scores obtained measures the overall evaluation of the quality of generated images, structural similarity to real post-operative images, and perceptual similarity. The results for the two approaches are labeled as result 1 and result 2 below.

5.1. Result-1:

The custom conditional GAN model (illustrated in Fig. 2, Fig. 3 and summarized in Table 2) demonstrates promising results in transforming pre-operative OCT images to their post-operative OCT scans. It achieved an average SSIM of 0.62, which suggests a moderate to high level of structural similarity. Furthermore, the model obtained the PSNR of 28.9, MSE of 0.001, and MAE of 0.02. These metrics suggest that the generated images are reasonably close to the real post-operative images in terms of pixel-wise differences. While these results are encouraging, further analysis and comparison with other methods are necessary to fully assess the model's capabilities. However, the model struggled with severely deformed OCT images, often failing to accurately reconstruct intricate structural details, which suggests potential areas for improvement in handling complex cases.

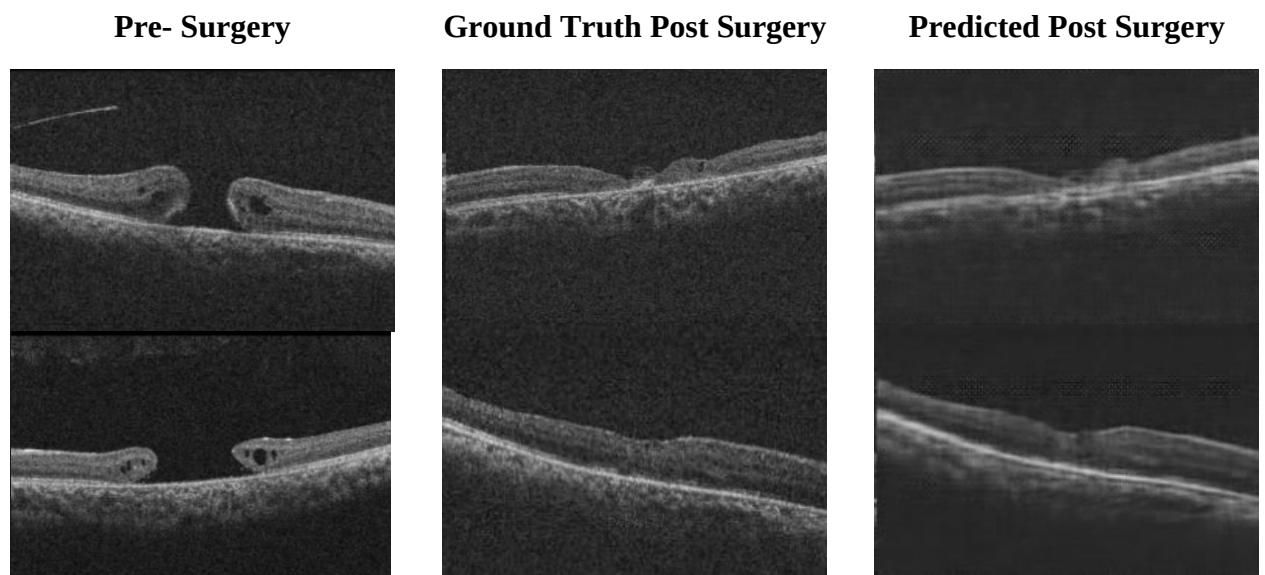


Fig.2 Generated result of Approach-1(Custom cGAN)

Table.2 Tabulated results of approach-1 (Custom cGAN)

SSIM	PSNR	MAE	MSE	LPIPS
0.662	28.227	0.023	0.001	0.312

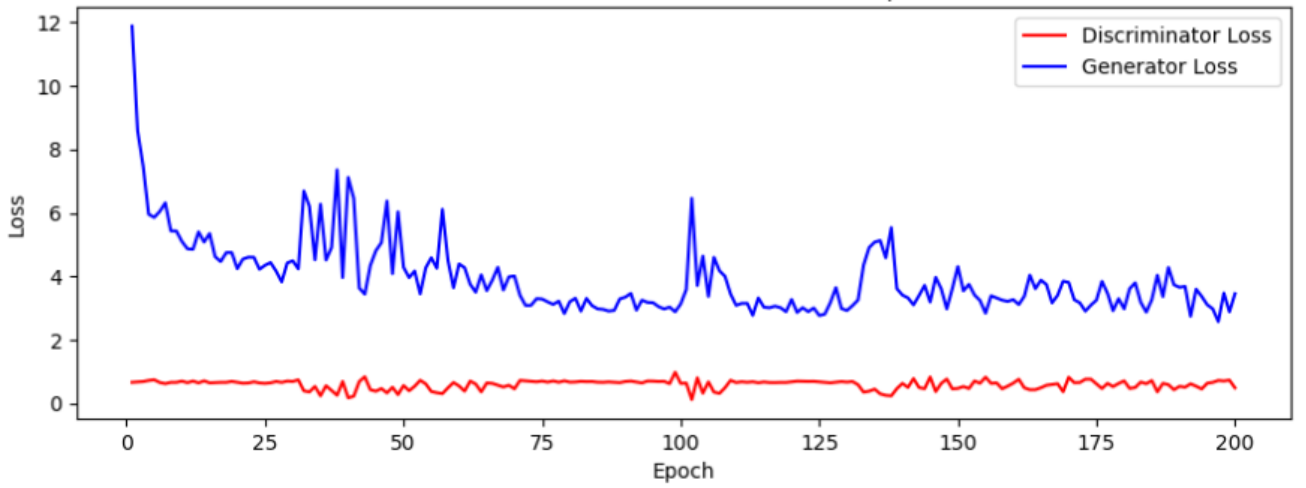


Fig.3 Discriminator loss and Generator loss over iterations of approach-1(Custom cGAN)

5.2. Result-2:

The modified Pix2Pix model (shown in Fig 4, Fig 5 and detailed in Table 3) exhibited even more promising results in predicting post-operative OCT morphology. The average SSIM of 0.557 demonstrates a similar level of structural similarity to the cGAN. The model with self attention[24] also achieved an MAE of 0.11, a PSNR of 26.01, and an LPIPS of 0.09. The comparable or better performance in other metrics suggest that the model is more effective at capturing the complex transformations between pre-operative and post-operative OCT images.

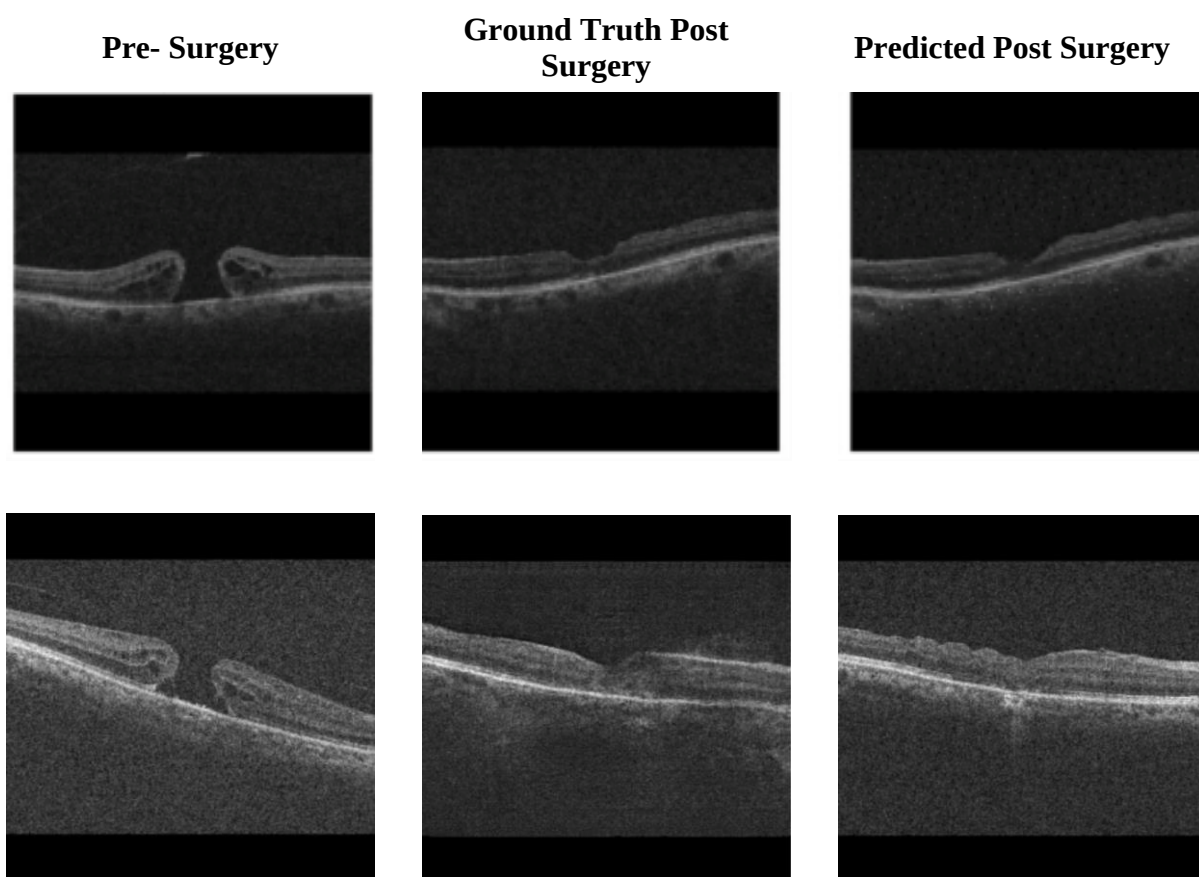


Fig.4 Generated results of approach-2(Modified Pix2Pix)

Table.3 Tabulated results of approach-2 (Modified Pix2Pix)

SSIM	PSNR	MAE	MSE	LPIPS
0.554	26.854	0.065	0.013	0.149

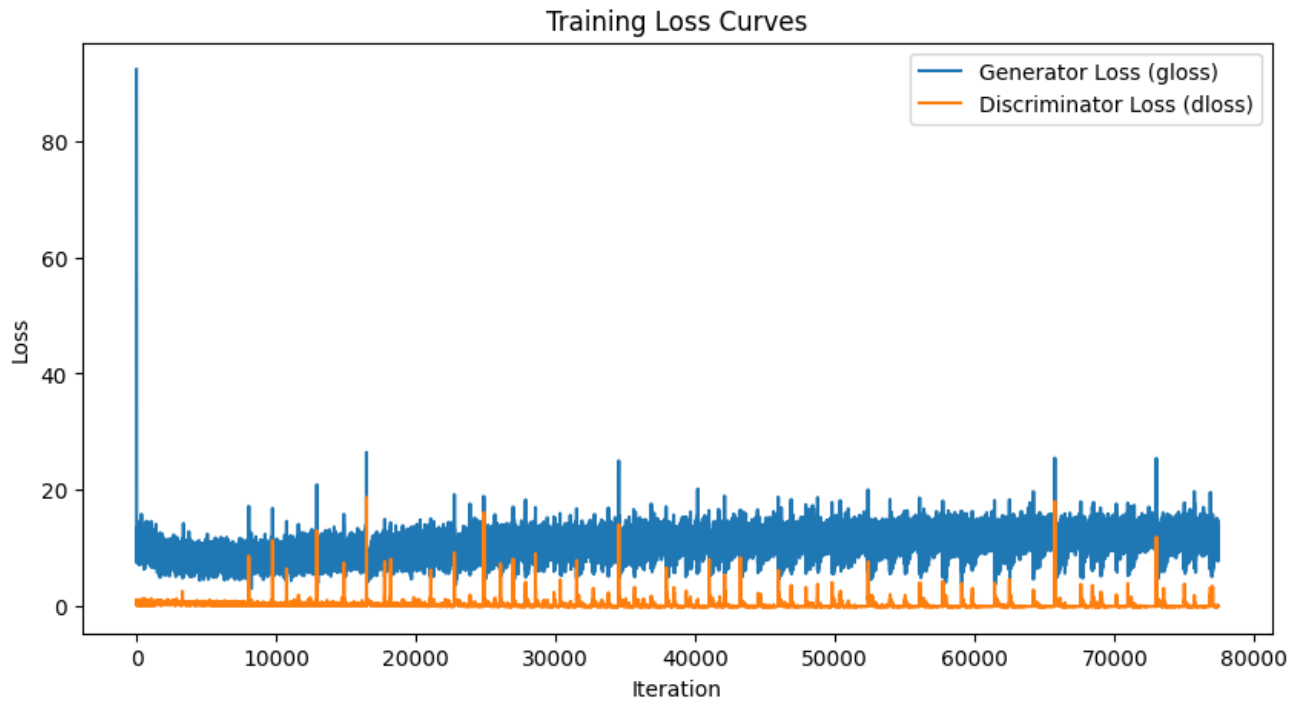


Fig. 5 Discriminator loss and Generator loss over iterations of approach-2(Modified Pix2Pix)

Experimental results demonstrate the capability of deep learning-driven image-to-image translation models, particularly the Pix2Pix architecture, for accurately predicting post-operative OCT morphology in MH surgery. The improved performance of the modified Pix2Pix model appears to be largely attributable to its well-established architecture. Its U-Net generator with skip connections, combined with the PatchGAN discriminator, effectively captures and preserves fine-grained structural details, thereby handling complex image transformations more adeptly than the custom cGAN. While additional training data can contribute to performance gains, the architectural design of modified Pix2Pix seems to play a more critical role in its enhanced predictive capability. The potential to predict post-operative OCT morphology offers a valuable tool for surgical planning and patient counseling. By visualizing the potential post-operative outcomes, surgeons and clinicians can make more informed decisions regarding the

surgical techniques. In this way the patients can also have a clearer understanding of the expected results.

6. Conclusion

This research study has been able to demonstrate the potential of Generative Adversarial Networks (GANs) in predicting post-operative macular hole (MH) closure from given pre-operative scans. We developed and compared two GAN models: a custom conditional GAN and a modified Pix2Pix model with self-attention to create synthetic post-operative OCT images. Our findings emphasize the ability of these models in identifying fine morphological changes involved in the closure of macular holes. Quantitative measurements, such as SSIM, PSNR, MSE, MAE and LPIPS show the excellence and clinical validity of the images produced and visual inspection also reinforces their capability for predicting post-operative retinal anatomy. With this study, we aim to add to the expanding literature for deep learning in ophthalmology, in particular for MH diagnosis and treatment. Precise post-operative estimation of pre-operative results has the potential to greatly improve treatment planning, patient counseling, and visual outcomes. By offering clinicians visualizations of possible post-operative outcomes, these models can facilitate informed decision-making for clinicians and patients alike. One of the strengths of our research is the explicit comparison of two GAN models (custom cGAN and modified Pix2Pix) for this purpose, with the insights gained into their respective strengths and limitations.

Specifically, our work shares the goal of predicting post-operative MH OCT images with the work presented in [1], which employed a VAE-based GDLM. While both studies aim to provide insights into post-operative macular features, differences exist in approach and datasets. Their study utilized a larger, augmented dataset (120,000 training, 5,000 validation pairs) from 150 patients, likely contributing to their reported high accuracy and F1 scores (94.7% and 0.891,

respectively), and strong agreement in foveolar height and mean foveal thickness measurements. Their focus on ELM and EZ recovery prediction accuracy (88.0% and 92.0%, respectively) highlights a specific clinical application. Our study, using a smaller dataset, explored cGANs, showing encouraging levels of similarity between generated and real post-operative images, particularly with the modified Pix2Pix model.

A potential shortcoming of our study is the smaller dataset, which could limit generalizability. Additionally, we did not specifically evaluate ELM and EZ recovery, focusing instead on overall image quality and structural similarity. Future work will address these limitations by expanding our dataset and incorporating specific evaluation of clinically relevant features like ELM and EZ integrity. We also plan to explore attention mechanisms, alternative loss functions, and develop a user-friendly clinical tool integrating these models. Rigorous clinical validation will be essential.

Furthermore, integrating this technology into clinical workflows requires careful consideration of ethical implications, data security, and standardization. Ensuring patient privacy, data integrity, and responsible AI use are paramount.

Despite these limitations, this work represents a major advancement toward applying generative deep learning for MH surgical outcome prediction. The development of accurate and reliable predictive tools offers clinicians a powerful way To aid in well-informed decisions, ultimately enhancing patient care and outcomes. Future research will refine the models, address limitations, and pave the way for clinical applications.

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